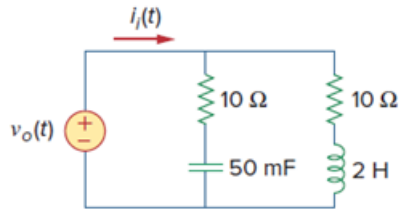


Problem

Find the transfer function $V_o(\omega)/I_i(\omega)$ for the circuit in Fig. 14.7. Obtain its zeros and poles.

Figure 14.7:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 14.7 in the textbook.

Capacitance in frequency domain is,

$$\begin{aligned} X_c &= \frac{1}{j\omega C} \\ &= \frac{1}{j\omega(50 \times 10^{-3})} \\ &= \frac{20}{j\omega} \end{aligned}$$

Inductance in frequency domain is,

$$\begin{aligned} X_L &= j\omega L \\ &= 2j\omega \end{aligned}$$

In the circuit, 10 Ω resistor and 50 mF capacitor are connected in series. Hence, the resultant impedance is,

$$Z_{eq1} = 10 + \frac{20}{j\omega}$$

Similarly, 10 Ω resistor and 2 H inductor are also connected in series. Hence, the resultant impedance is,

$$Z_{eq2} = 10 + 2j\omega$$

Step 2 of 3

The total impedance of the circuit is the transfer function of the circuit.

Calculate the transfer function.

$$\begin{aligned}\frac{V_o(j\omega)}{I_i(j\omega)} &= Z_{eq1} \parallel Z_{eq2} \\ &= \frac{\left(10 + \frac{20}{j\omega}\right)(10 + 2j\omega)}{10 + \frac{20}{j\omega} + 10 + 2j\omega} \\ &= \frac{(10j\omega + 20)(10 + 2j\omega)}{(10j\omega + 20) + j\omega(10 + 2j\omega)}\end{aligned}$$

Substitute $j\omega$ for s .

$$\begin{aligned}\frac{V_o(s)}{I_i(s)} &= \frac{(10s + 20)(10 + 2s)}{(10s + 20) + s(10 + 2s)} \\ &= \frac{(10)(s + 2)(2)(s + 5)}{10s + 20 + 10s + 2s^2} \\ &= \frac{(20)(s + 2)(s + 5)}{2s^2 + 20s + 20} \\ &= \frac{10(s + 2)(s + 5)}{s^2 + 10s + 10}\end{aligned}$$

Therefore, the transfer function is $\boxed{\frac{10(s + 2)(s + 5)}{s^2 + 10s + 10}}$.

[Comments \(2\)](#)



Anonymous

why is it $j\omega(10+2j\omega)$ and not just $10+2j\omega$ in the denominator to the right? where did the extra " $j\omega$ " come from?



Anonymous

wait , I see, but why did they only multiply by $j\omega$ throughout both terms in the denominator but not numerator

Zeros of the transfer function are obtained by equating the numerator of the transfer function to zero.

$$10(s+2)(s+5) = 0$$

$$(s+2)(s+5) = 0$$

$$s_{1,2} = -2, -5$$

Therefore, the zeros of the transfer function are $\boxed{-2}$ and $\boxed{-5}$.

The poles of the transfer function are obtained by equate the denominator of the transfer function to zero.

$$s^2 + 10s + 10 = 0$$

$$s_{1,2} = \frac{-10 \pm \sqrt{(10)^2 - 4(1)(10)}}{2}$$

$$= \frac{-10 \pm \sqrt{100 - 40}}{2}$$

$$= \frac{-10 \pm 7.746}{2}$$

$$= -1.127, -8.873$$

Therefore, the poles of the transfer function are $\boxed{-1.127}$ and $\boxed{-8.873}$.